Capstone Development Reflection Journal

Module B: AI for Leaders

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Week 2

Assignment/Capstone work: I identified my industries of interest: finance, environment, and entertainment. Finance because I am a valuation specialist at a hedge fund in Hong Kong, so valuing derivatives is my profession and my deepest interest; environment because the EV transition is the most consequential industrial shift of my lifetime; entertainment because music and film are domains I know as a consumer.

Reflection: Narrowing to three industries was the first real decision. My instinct was to chase the most technically impressive datasets, but the industries-first framing forced a better question: which problems would I still care about after the grade is posted? Markets, the energy transition, and culture. Everything since has been downstream of that choice.

Week 3

Assignment/Capstone work: I extracted the first real data: the SPY options snapshot for December 31, 2024 (9,646 contracts, 23 columns) and the CPCA China auto file (38,806 model-month rows). Problems spotted immediately: 625 zero-bid option contracts (dead quotes), duplicate model-month rows, and “NA” strings in the body-type column.

Reflection: Data quality is not a footnote, it is the analysis. The zero-bid contracts looked ordinary until I asked what a zero bid means: nobody will buy at any price. Deleting 6.5% of the data before modeling felt drastic, but training on prices that do not exist is worse.

Week 4

Assignment/Capstone work: For each dataset I specified 1–2 machine learning types with justification — regression (SPY IV, TMDB revenue, FINRA short shares from trading volume), classification (China EV flag, Spotify popularity, FINRA crowded-stock flag), clustering (Spotify archetypes) — plus 2–3 research questions per dataset checked against the rubric.

Reflection: The module's framing of predictive modeling as supervised segmentation reframed how I see market data: an option chain is a segmentation problem where the model learns boundaries across moneyness and expiry the way a tree segments customers. The industry boundary matters less than the problem structure.

Week 5

Assignment/Capstone work: I finalized the five datasets: SPY options (finance), China EV sales (environment), Spotify tracks and TMDB movies (entertainment), and a FINRA equity short-interest file for S&P 500 stocks (finance) for the fifth slot. With Kaggle unreachable from my machine, Spotify and TMDB were sourced from their Kaggle-published originals via local copies. Confirmed: three industries, mixed types, two-plus learning types, genuine data-quality problems in all five.

Reflection: Dataset selection is mostly subtraction. Letting go of the CO2 panel hurt — the data was beautiful — but the rubric rewards fit over volume: five datasets I can defend beat eight I cannot.

Week 6

Assignment/Capstone work: I audited the three new datasets — 450 duplicate rows and 24,259 duplicate track ids (label noise) in Spotify; 17 duplicates and zero-coded budgets in TMDB; corporate-action jumps, a reused ticker, and uneven coverage in the FINRA short-interest file —

and restructured the paper per-section with the rubric's exact labels, double spacing, and one page of examples.

Reflection: The Spotify label noise was the week's best puzzle: identical features with contradictory genre labels breaks naive classification, and the fix changes the problem statement. Formatting discipline was the unglamorous lesson: double spacing halves words per page, so every sentence must earn its place.